Locomotion and Movement

Movement is one of the significant features of living beings. Animals and plants exhibit a wide range of movements. The **streaming of protoplasm** in unicellular organisms like *Amoeba* is a simple form of movement; the movement of **cilia, flagella and tentacles** is shown by many organisms; and human beings can move their **limbs, jaws, eyelids, tongue,** etc.

- **Locomotion -** some movements result in a change of place or location. Such voluntary movements are called locomotion. Walking, running, climbing, flying and swimming are all forms of locomotory movements.
  - Locomotory structures **need not be different** from those affecting other types of movements. For example, in *Paramoecium*, cilia help in the movement of food through the cytopharynx *and* in locomotion as well; *Hydra* can use its tentacles for capturing prey *and* for locomotion; and we use our limbs for changes in body posture *and* for locomotion.

These observations suggest that movements and locomotion **cannot be studied separately**. The two may be linked by stating that **all locomotions are movements but all movements are not locomotions**.

- Methods of locomotion vary with the animal's **habitat** and the demand of the situation. However, locomotion is **generally** for the search of food, shelter, mate, suitable breeding grounds, favourable climatic conditions, or to escape from enemies/predators.

17.1 Types of Movement

Cells of the human body exhibit **three main types of movements**, namely, **amoeboid, ciliary and muscular**.

Type	Where it occurs	How it works
Amoeboid	Specialised cells like **macrophages** and **leucocytes** in blood	Effected by **pseudopodia** formed by the streaming of protoplasm (as in *Amoeba*); **cytoskeletal elements like microfilaments** are also involved.
Ciliary	Most internal tubular organs lined by **ciliated epithelium**	Coordinated cilia in the **trachea** help remove dust particles and foreign substances inhaled with air; the passage of **ova** through the female reproductive tract is also facilitated by ciliary movement.
Muscular	Limbs, jaws, tongue, etc.	Uses the **contractile property of muscles**; effectively used for locomotion and other movements by human beings and the **majority** of multicellular organisms.

> ⓘ ***[NOTE]** Locomotion requires a **perfect coordinated activity of the muscular, skeletal and neural systems**. This chapter covers the types of muscles, their structure, the mechanism of their contraction, and the important aspects of the skeletal system.*

17.2 Muscle

You have studied in Chapter 8 that **cilia and flagella are outgrowths of the cell membrane**. Flagellar movement helps in the **swimming of spermatozoa**, the maintenance of water current in the **canal system of sponges**, and in the locomotion of Protists like *Euglena*.

- **Muscle -** a specialised tissue of **mesodermal origin**. About **40-50 per cent** of the body weight of a human adult is contributed by muscles.
  - Muscles have special properties: **excitability, contractility, extensibility and elasticity**.

Muscles have been classified using different criteria, namely **location, appearance and nature of regulation** of their activities. Based on their **location**, three types are identified: **(i) Skeletal (ii) Visceral** and **(iii) Cardiac**.

Muscle type	Location	Appearance	Regulation	Chief role
Skeletal	Closely associated with the **skeletal components** of the body	**Striped / striated** under the microscope	Under **voluntary** control of the nervous system (voluntary muscles)	Primarily **locomotory actions** and changes of body posture

Muscle type	Location	Appearance	Regulation	Chief role
Visceral	Inner walls of hollow visceral organs - alimentary canal, reproductive tract, etc.	**No striation, smooth** (smooth / nonstriated muscle)	**Not** under voluntary control (involuntary muscles)	Transport of food through the digestive tract and gametes through the genital tract
Cardiac	Muscles of the **heart**; many cells assemble in a **branching pattern**	**Striated**	**Involuntary** - the nervous system does not control them directly	Contraction of the heart

Structure of a skeletal muscle. Each organised skeletal muscle is made of a number of **muscle bundles** or **fascicles** held together by a common collagenous connective tissue layer called **fascia**. Each muscle bundle contains a number of **muscle fibres** (Figure 17.1). The bundle is also served by a **blood capillary** supply drawn in the cross-section.

- **Sarcolemma / sarcoplasm.** Each muscle fibre is lined by the plasma membrane called **sarcolemma** enclosing the **sarcoplasm**. A muscle fibre is a **syncitium** (source spelling) as the sarcoplasm contains **many nuclei**.
  - The endoplasmic reticulum, i.e. the **sarcoplasmic reticulum** of the muscle fibres, is the **store house of calcium ions** (Ca^{++}).
  - A characteristic feature of the muscle fibre is a large number of parallelly arranged **filaments** in the sarcoplasm called **myofilaments** or **myofibrils**.

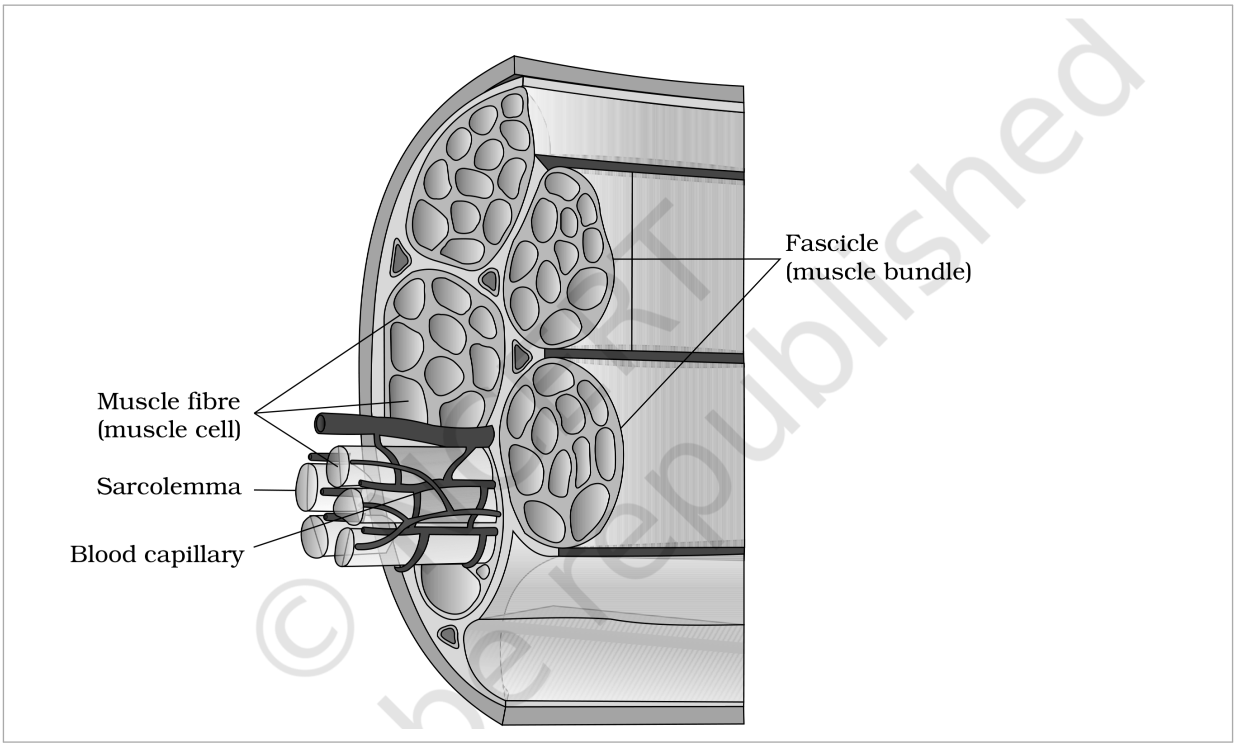

Figure 17.1 Diagrammatic cross sectional view of a muscle showing muscle bundles and muscle fibres

> ⓘ ***[NOTE] Figure 17.1 labels:*** *Fascicle (muscle bundle); Muscle fibre (muscle cell); Sarcolemma; Blood capillary.*

Banding pattern. Each myofibril has **alternate dark and light bands** on it. The striated appearance is due to the distribution pattern of two important proteins - **Actin and Myosin**.

Band / line	Also called	Contents
'I' band (light)	Isotropic band	Contains **actin** (thin filament)
'A' band (dark)	Anisotropic band	Contains **myosin** (thick filament)
'Z' line	Elastic fibre bisecting each 'I' band	Thin filaments are **firmly attached** to it
'M' line	Thin fibrous membrane	Holds the thick filaments together in the **middle of the 'A' band**
'H' zone	Central part of the 'A' band	The part of the thick filament **not overlapped** by thin filaments - i.e. **thick filaments only**

- Both proteins are arranged as **rod-like structures**, parallel to each other and to the longitudinal axis of the myofibrils. **Actin filaments are thinner** than the myosin filaments, hence they are commonly called **thin** and **thick** filaments respectively.
- The **'A' and 'I' bands are arranged alternately** throughout the length of the myofibrils.

● **Sarcomere -** the portion of the myofibril between **two successive 'Z' lines**. It is the **functional unit of contraction** (Figure 17.2). (The muscle **fibre** itself is the **anatomical unit** of muscle, while the sarcomere is its functional unit.)

- In a **resting state**, the edges of the thin filaments on either side of the thick filaments **partially overlap** the free ends of the thick filaments, leaving the central part of the thick filaments (the **'H' zone**) free.

☆ ***[MEMORY AID - not in NCERT]*** *Each **sarcomere** = one central **'A' band** (thick myosin filaments) flanked by **two half 'I' bands** (thin actin filaments), the whole unit marked off by **'Z' lines** at both ends. Mnemonic for what shortens: only the **'I'** band and the **'H'** zone shrink on contraction - the **'A'** band never does.*

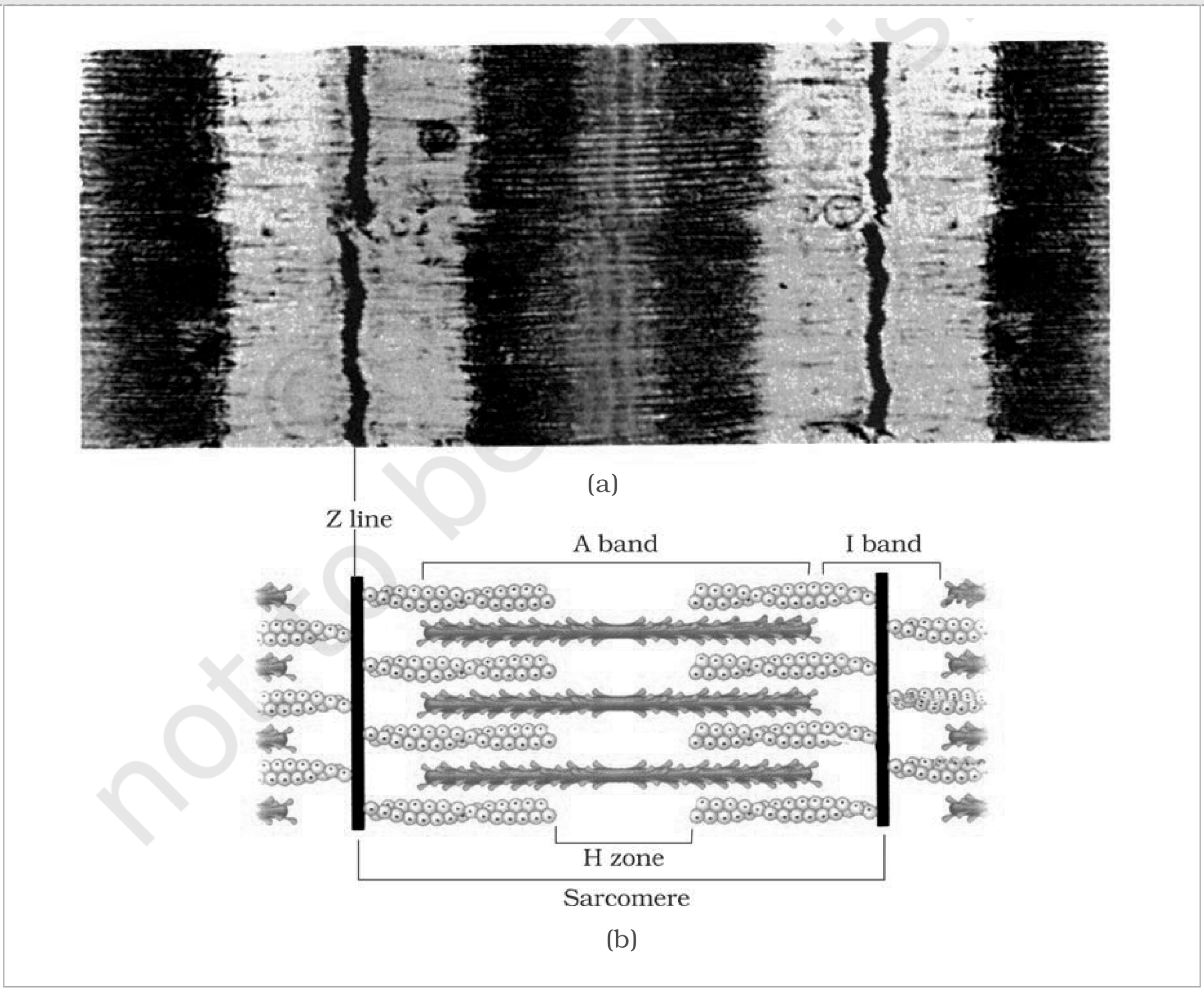

Figure 17.2 Diagrammatic representation of (a) anatomy of a muscle fibre showing a sarcomere (b) a sarcomere

ⓘ ***[NOTE] Figure 17.2 labels:*** *Z line; A band; I band; H zone; Sarcomere; panels (a) and (b).*

17.2.1 Structure of Contractile Proteins

Actin (thin) filament. Each actin filament is made of two **'F' (filamentous) actins** helically wound to each other. Each 'F' actin is a polymer of monomeric **'G' (Globular) actins**.

- Two filaments of another protein, **tropomyosin**, also run close to the 'F' actins throughout their length.
- A complex protein **Troponin** is distributed at regular intervals on the tropomyosin. In the **resting state**, a subunit of troponin **masks the active binding sites for myosin** on the actin filaments (Figure 17.3a).

Myosin (thick) filament. Each myosin filament is also a polymerised protein. Many monomeric proteins called **Meromyosins** (Figure 17.3b) constitute one thick filament.

Part of a meromyosin	Name	Feature
Globular **head** + short arm	**Heavy meromyosin (HMM)**	Projects outwards at regular distance and angle from the surface of the filament, forming the **cross arm**; the head is an **active ATPase enzyme** with binding sites for **ATP** and active sites for **actin**.
Tail	**Light meromyosin (LMM)**	The rod-like shaft of the monomer.

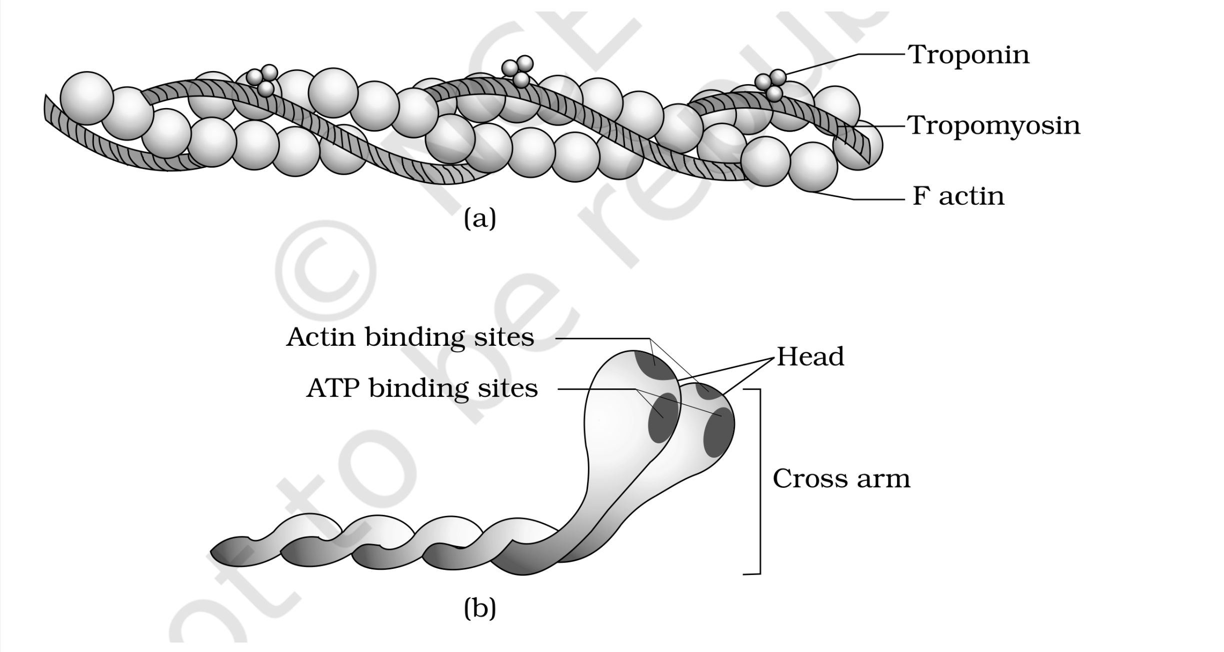

Figure 17.3 (a) An actin (thin) filament (b) Myosin monomer (Meromyosin)

> ⓘ ***[NOTE] Figure 17.3 labels:*** *Troponin; Tropomyosin; F actin; Actin binding sites; ATP binding sites; Head; Cross arm; panels (a) and (b).*

17.2.2 Mechanism of Muscle Contraction

- **Sliding filament theory -** the mechanism of muscle contraction is best explained by this theory, which states that contraction of a muscle fibre takes place by the **sliding of the thin filaments over the thick filaments**.
- **Motor unit -** a motor neuron along with the muscle fibres connected to it. The junction between a motor neuron and the sarcolemma of the muscle fibre is the **neuromuscular junction** or **motor-end plate**.

The contraction cycle, step by step:

1. Contraction is initiated by a signal sent by the **central nervous system (CNS)** via a **motor neuron**.
2. The neural signal reaching the neuromuscular junction releases a neurotransmitter (**Acetyl choline**), which generates an **action potential** in the sarcolemma.
3. The action potential spreads through the muscle fibre and causes the release of **calcium ions (Ca^{++})** into the sarcoplasm.
4. The rise in Ca^{++} level leads to the binding of calcium with a subunit of **troponin** on the actin filaments, which **removes the masking** of the active sites for myosin.
5. Using energy from **ATP hydrolysis**, the **myosin head** binds to the exposed active sites on actin to form a **cross bridge** (Figure 17.4).
6. The cross bridge **pulls the attached actin filaments towards the centre of the 'A' band**; the 'Z' lines attached to these actins are also pulled inwards, **shortening the sarcomere -** i.e. contraction.
7. The myosin head, releasing the **ADP and P_i**, goes back to its relaxed state. A **new ATP** binds and the cross bridge is **broken** (Figure 17.4).

8 The ATP is again hydrolysed by the myosin head and the cycle of cross-bridge formation and breakage repeats, causing further **sliding**.

9 The process continues till the Ca^{++} ions are **pumped back** into the sarcoplasmic cisternae, masking the actin filaments again; the 'Z' lines return to their original position - i.e. **relaxation**.

- During shortening (contraction), the **'I' bands get reduced**, whereas the **'A' bands retain their length** (Figure 17.5). The figure stages this across three states of the sarcomere: **Relaxed**, **Contracting** and **Maximally Contracted**.
- The **reaction time** of the fibres can vary in different muscles.

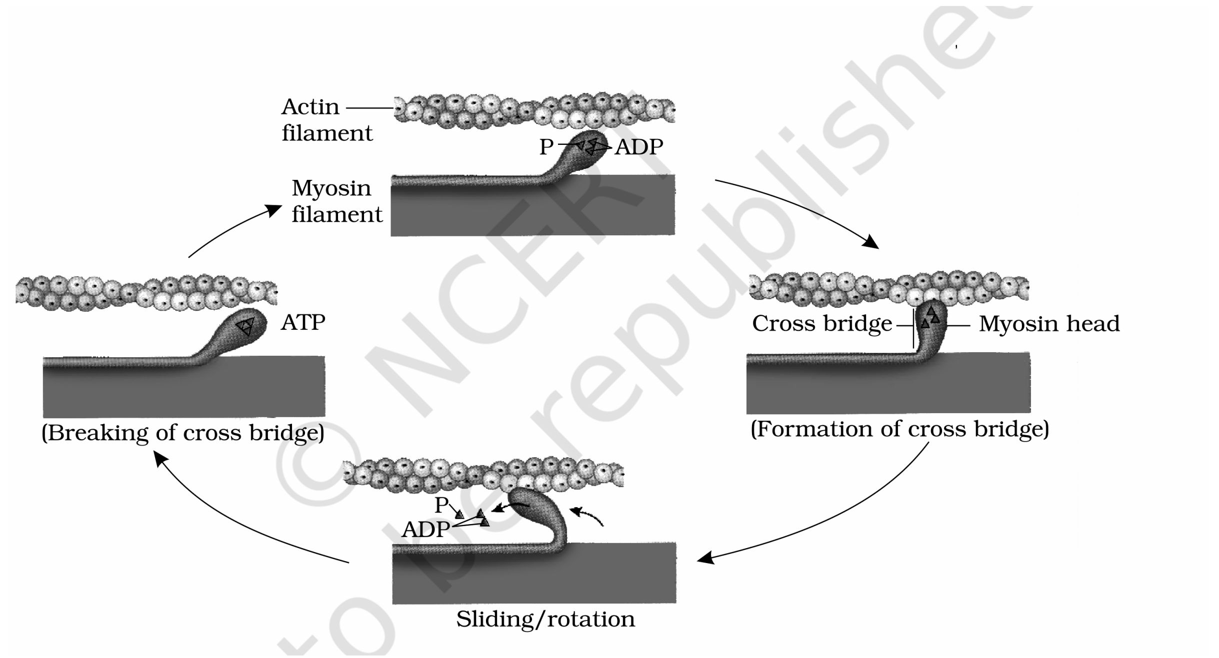

Figure 17.4 Stages in cross bridge formation, rotation of head and breaking of cross bridge

> ⓘ ***[NOTE] Figure 17.4 labels:*** *Actin filament; Myosin filament; P; ADP; ATP; Cross bridge; Myosin head; Sliding/rotation of the head; (Formation of cross bridge); (Breaking of cross bridge).*

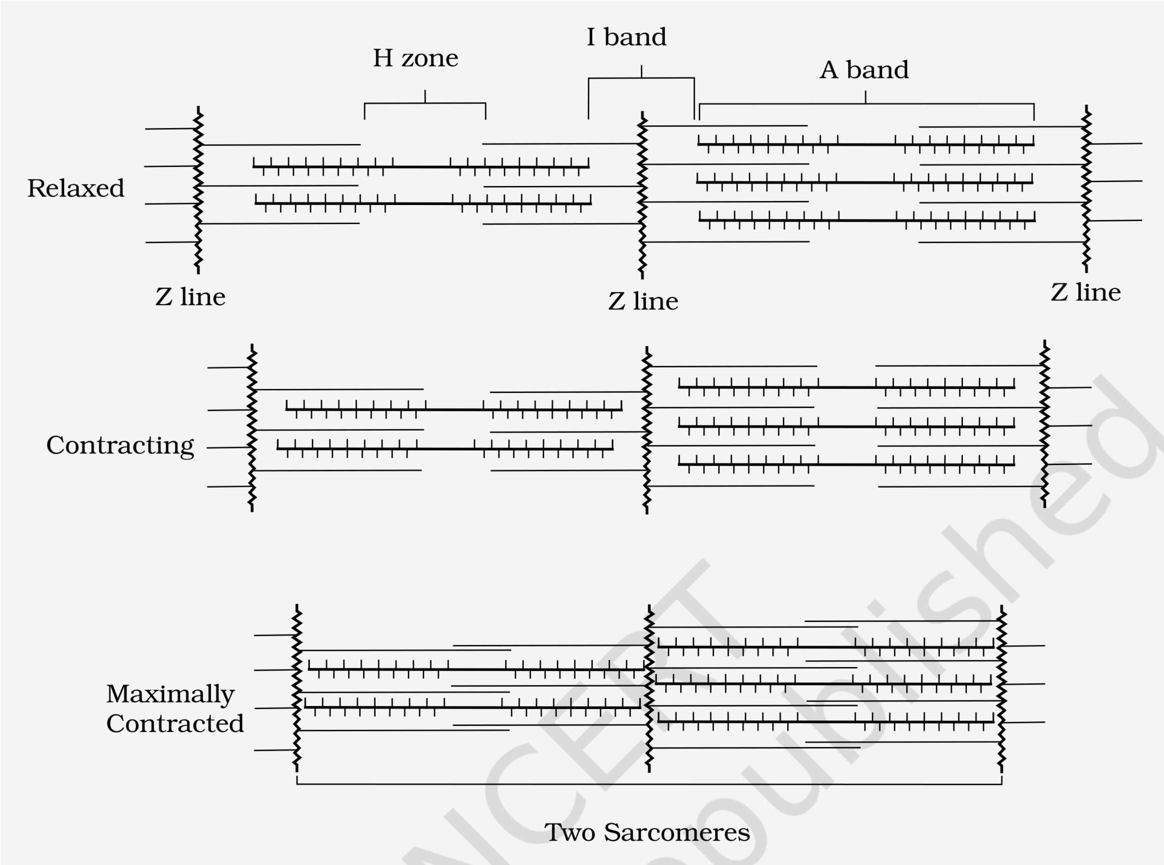

Figure 17.5 Sliding-filament theory of muscle contraction (movement of the thin filaments and the relative size of the I band and H zones)

> ⓘ ***[NOTE] Figure 17.5 labels:*** *H zone; I band; A band; Z line; Two Sarcomeres; and the three staged states Relaxed, Contracting and Maximally Contracted.*

Fatigue. Repeated activation of the muscles can lead to the accumulation of **lactic acid** due to the **anaerobic breakdown of glycogen** in them, causing **fatigue**.

Red vs White fibres. Muscles are classified as Red and White fibres based primarily on the amount of the red-coloured oxygen-storing pigment **myoglobin**.

Feature	Red fibres	White fibres
Myoglobin	**High** - gives a reddish appearance	**Low** - appear pale / whitish
Mitochondria	**Plenty**	**Few**
Sarcoplasmic reticulum	Lower	**High**
Energy source	Use stored O_2 for ATP - **aerobic muscles**	Depend on the **anaerobic** process for energy

17.3 Skeletal System

The **skeletal system** consists of a framework of **bones and a few cartilages**. This system has a significant role in the movement shown by the body - imagine chewing food without jaw bones and walking around without limb bones.

- **Bone and cartilage** are specialised **connective tissues**. Bone has a very **hard matrix** due to **calcium salts**; cartilage has a slightly **pliable matrix** due to **chondroitin salts**.
- In human beings, the skeletal system is made up of **206 bones** and a few cartilages, grouped into two principal divisions - the **axial** and the **appendicular** skeleton.

17.3.axial Axial Skeleton

The **axial skeleton** comprises **80 bones** distributed along the main axis of the body: the **skull, vertebral column, sternum and ribs**.

Skull (Figure 17.6). The skull is composed of two sets of bones **- cranial and facial -** that total **22 bones**. Cranial bones are **8** in number and form the hard protective outer covering, the **cranium**, for the brain. The facial region is made up of **14** skeletal elements that form the front part of the skull.

- A single **U-shaped bone called hyoid** is present at the base of the buccal cavity.
- Each middle ear contains **three tiny bones - Malleus, Incus and Stapes**, collectively called **Ear Ossicles**.
- The skull articulates with the superior region of the vertebral column with the help of **two occipital condyles (dicondylic skull)**.

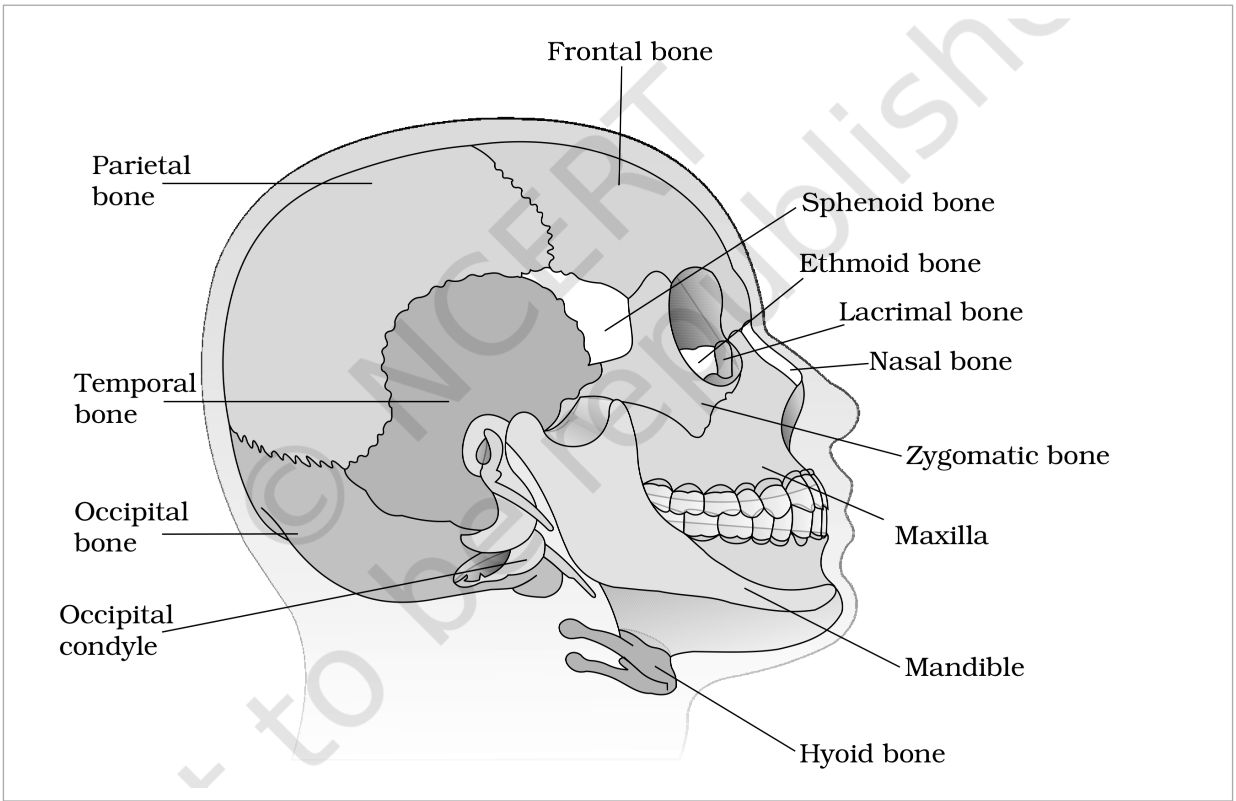

Figure 17.6 Diagrammatic view of human skull

> ⓘ ***[NOTE] Figure 17.6 labels:*** *Parietal bone; Frontal bone; Temporal bone; Occipital bone; Occipital condyle; Sphenoid bone; Ethmoid bone; Lacrimal bone; Nasal bone; Zygomatic bone; Maxilla; Mandible; Hyoid bone.*

Vertebral column (Figure 17.7) is formed by **26 serially arranged units called vertebrae** and is dorsally placed. It extends from the base of the skull and constitutes the main framework of the trunk.

- Each vertebra has a central hollow portion (**neural canal**) through which the **spinal cord** passes. The first vertebra is the **atlas**, and it articulates with the occipital condyles.
- The column is differentiated into **cervical (7), thoracic (12), lumbar (5), sacral (1-fused) and coccygeal (1-fused)** regions, starting from the skull. The number of **cervical vertebrae is seven in almost all mammals**, including human beings. (Adjacent vertebrae are separated by an **intervertebral disc** of cartilage, which forms the cartilaginous joint discussed in 17.4.)
- The vertebral column **protects the spinal cord, supports the head**, and serves as the point of attachment for the ribs and the musculature of the back.

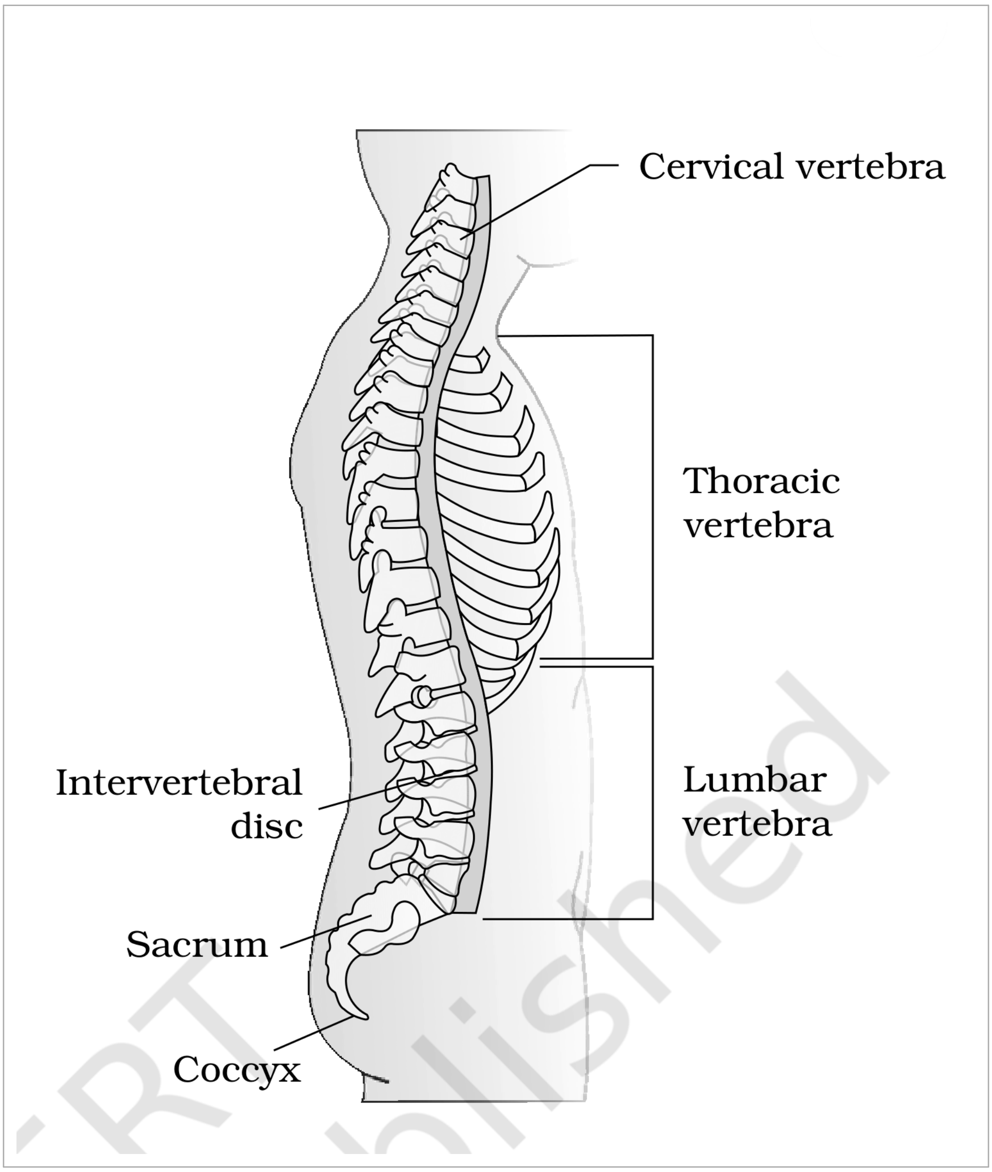

Figure 17.7 Vertebral column (right lateral view)

> ⓘ ***[NOTE] Figure 17.7 labels:*** *Cervical vertebra; Thoracic vertebra; Lumbar vertebra; Intervertebral disc; Sacrum; Coccyx.*

Sternum and ribs. The **sternum** is a flat bone on the ventral midline of the thorax. There are **12 pairs of ribs**. Each rib is a thin flat bone connected dorsally to the vertebral column and ventrally to the sternum; it has **two articulation surfaces** on its dorsal end and is hence called **bicephalic**.

Rib set	Pairs	Attachment
True ribs	First **7** pairs	Dorsally attached to the thoracic vertebrae; ventrally connected to the sternum with **hyaline cartilage**
Vertebrochondral (false) ribs	**8th, 9th, 10th** pairs	Do **not** articulate directly with the sternum; join the **seventh rib** with hyaline cartilage
Floating ribs	Last **2** pairs (11th, 12th)	**Not connected ventrally**

- Thoracic vertebrae, ribs and sternum together form the **rib cage** (Figure 17.8).

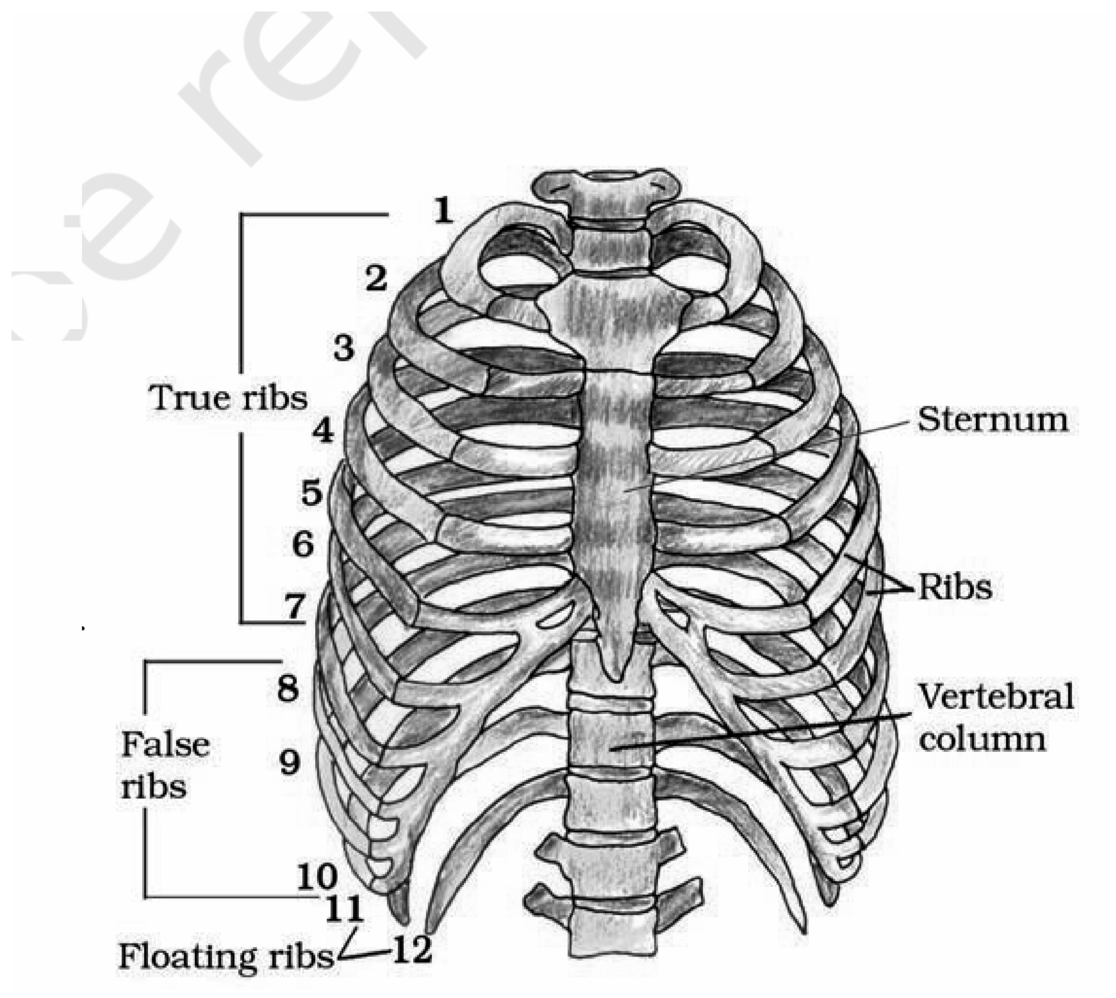

Figure 17.8 Ribs and rib cage

> ⓘ ***[NOTE] Figure 17.8 labels:*** *rib pairs numbered 1 to 12; True ribs; False ribs; Floating ribs; Sternum; Ribs; Vertebral column.*

17.3.append

Appendicular Skeleton

- **Appendicular skeleton** - the bones of the limbs along with their girdles. Each limb is made of **30 bones**.

Limb	Bones (with counts)
Hand / fore limb (Figure 17.9)	Humerus, radius and ulna, **carpals** (wrist bones - **8**), **metacarpals** (palm bones - **5**) and **phalanges** (digits - **14**).
Leg / hind limb (Figure 17.10)	**Femur** (thigh bone - the **longest bone**), tibia and fibula, **tarsals** (ankle bones - **7**), **metatarsals** (**5**) and **phalanges** (digits - **14**).

- A cup-shaped bone called **patella** covers the knee ventrally (the **knee cap**).
- **Pectoral** and **Pelvic** girdle bones help in the articulation of the upper and lower limbs respectively with the axial skeleton. Each girdle is formed of two halves.

Pectoral girdle (Figure 17.9):

- Each half consists of a **clavicle** and a **scapula**.
- **Scapula** is a large **triangular flat bone** situated in the dorsal part of the thorax between the **second and the seventh ribs**. Its dorsal, flat, triangular body has a slightly elevated ridge called the **spine**, which projects as a flat, expanded process called the **acromion**; the clavicle articulates with this.
- Below the acromion is a depression called the **glenoid cavity**, which articulates with the **head of the humerus** to form the **shoulder joint**.
- Each **clavicle** is a long slender bone with two curvatures, commonly called the **collar bone**.

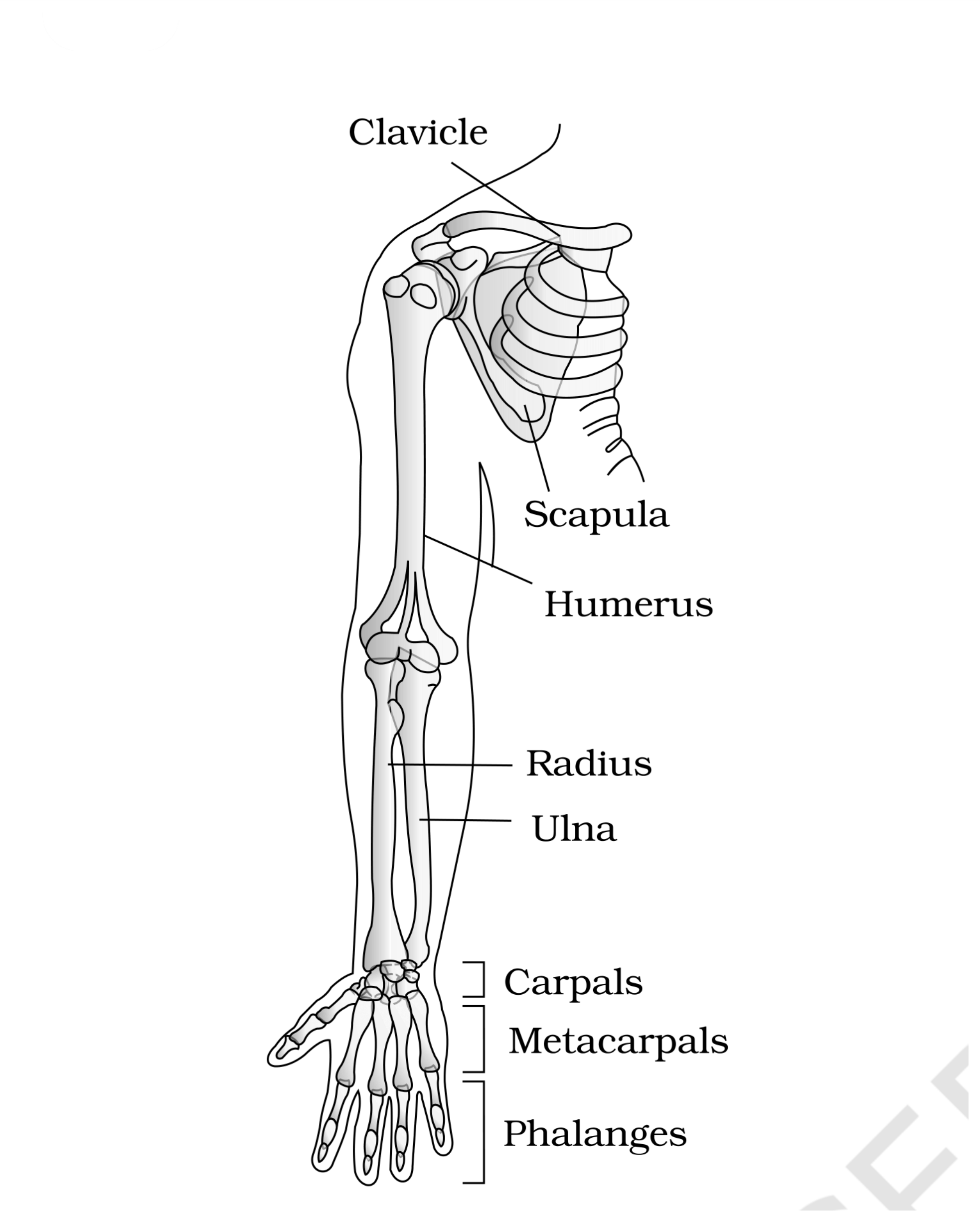

Figure 17.9 Right pectoral girdle and upper arm. (frontal view)

> ⓘ ***[NOTE] Figure 17.9 labels:*** *Clavicle; Scapula; Humerus; Radius; Ulna; Carpals; Metacarpals; Phalanges.*

Pelvic girdle (Figure 17.10):

- Consists of **two coxal bones**. Each coxal bone is formed by the fusion of **three bones - ilium, ischium and pubis**.
- At the point of fusion is a cavity called the **acetabulum**, to which the thigh bone (femur) articulates.
- The two halves of the pelvic girdle meet ventrally to form the **pubic symphysis**, containing fibrous cartilage.

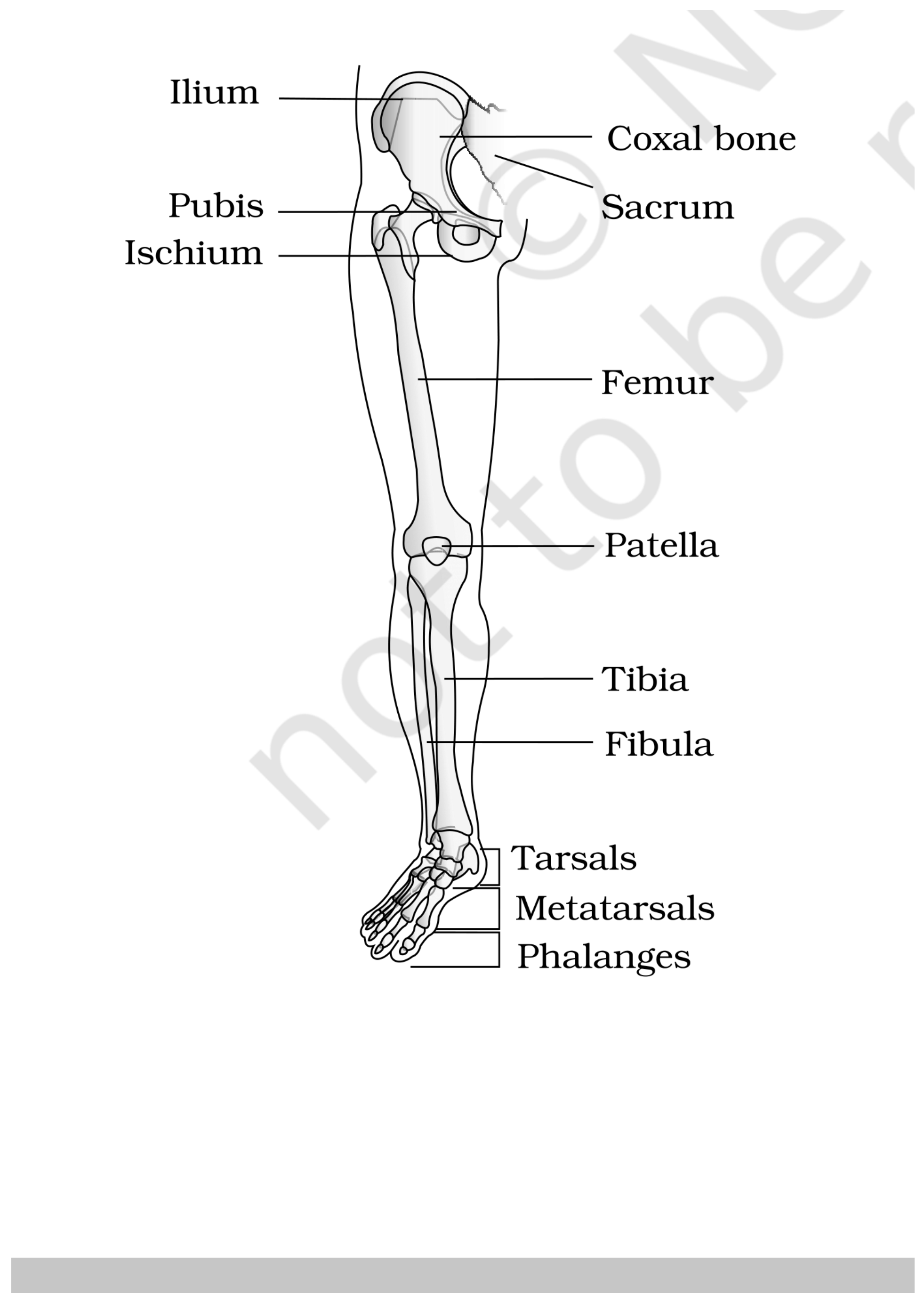

Figure 17.10 Right pelvic girdle and lower limb bones (frontal view)

> ⓘ ***[NOTE] Figure 17.10 labels:*** *Ilium; Pubis; Ischium; Coxal bone; Sacrum; Femur; Patella; Tibia; Fibula; Tarsals; Metatarsals; Phalanges.*

> ☆ ***[MEMORY AID - not in NCERT] Pectoral vs Pelvic girdle*** *at a glance: pectoral =* ***clavicle*** *+* ***scapula*** *(articulates the arm at the glenoid cavity/shoulder); pelvic = one* ***coxal bone*** *per half =* ***ilium*** *+* ***ischium*** *+* ***pubis*** *fused (articulates the thigh at the acetabulum).*

17.4 Joints

Joints are essential for all types of movements involving the bony parts of the body; locomotory movements are no exception.

- **Joints -** points of contact between bones, or between bones and cartilages. The force generated by the muscles is used to carry out movement through joints, where the joint acts as a **fulcrum**. The movability at these joints varies depending on different factors.

Joints have been classified into **three major structural forms**, namely, **fibrous, cartilaginous and synovial**.

Joint type	Movement	Example
Fibrous	Do **not allow any movement**	Flat **skull bones** fusing end-to-end via dense fibrous connective tissue as **sutures** to form the cranium
Cartilaginous	Permit **limited** movements	Joint between **adjacent vertebrae** in the vertebral column
Synovial	Allow **considerable** movement; help in locomotion and many other movements	Characterised by a fluid-filled **synovial cavity** between the articulating surfaces of the two bones

Synovial joints are characterised by the presence of a **fluid-filled synovial cavity** between the articulating surfaces of the two bones. Such an **arrangement** allows considerable movement. Sub-types with examples:

- **Ball and socket joint -** between humerus and pectoral girdle.
- **Hinge joint -** knee joint.
- **Pivot joint -** between atlas and axis.
- **Gliding joint -** between the carpals.
- **Saddle joint -** between carpal and metacarpal of the thumb.

17.5 Disorders of Muscular and Skeletal System

Disorder	Description
Myasthenia gravis	**Auto immune disorder** affecting the **neuromuscular junction**, leading to fatigue, weakening and paralysis of skeletal muscle.
Muscular dystrophy	**Progressive degeneration** of skeletal muscle, mostly due to a **genetic disorder**.
Tetany	Rapid spasms (wild contractions) in muscle due to **low** Ca^{++} in body fluid.
Arthritis	**Inflammation of joints**.
Osteoporosis	**Age-related** disorder characterised by **decreased bone mass** and increased chances of fractures. **Decreased levels of estrogen** is a common cause.
Gout	Inflammation of joints due to the accumulation of **uric acid crystals**.

> ***[NOTE]*** *Chapter map recap:* ***movement*** *(amoeboid / ciliary / muscular) leads to* ***muscle*** *(skeletal / visceral / cardiac), whose contractile unit is the* ***sarcomere****; contraction runs by the* ***sliding filament theory****; muscles act on the* ***skeletal system*** *(206 bones: axial + appendicular) through* ***joints****. Disorders can strike either the muscular side (myasthenia gravis, muscular dystrophy, tetany) or the skeletal/joint side (arthritis, osteoporosis, gout).*